Электроника МК-59

Desktop calculator: keys, operation and electronics. A reference sheet compiled from the operating manual (руководство по эксплуатации), the circuit diagram and the chip datasheets.

1. About the calculator

A 16-digit desktop calculator for planning, statistical and accounting work, made at the Positron / Rodon plant in Ivano-Frankivsk (Ukrainian SSR) from 1985. It has a 16-digit vacuum-fluorescent display, reed-switch keys, one memory register and mains power (220 V, about 8 W). Numbers are entered with a floating (“natural”) decimal point; results can also be shown with a fixed number of decimals (0–9), with rounding. The numbers 1–16 under the display mark the digit positions. Russian uses a comma as the decimal separator.

Two key-legend versions exist. The original manual is printed for the earlier legends; later calculators (and the circuit diagram) use the newer ones. The keys are in the same places:

Later calculators	Manual	Function
РФЕ (top tall key)	Ф	Fixed number of decimals
РЕЖ (bottom tall key)	Е	Normal (floating) decimal point
СХ	СК	Clear the number on the display
П→Х	ИП	Recall memory (memory kept)

2. Key reference

Key	Meaning	What it does
0–9 and ,	digits, decimal comma	Enter numbers.
+ − × ÷	arithmetic	Each key first completes any pending operation, then sets up the new one.
=	equals	Completes the pending operation. Pressing again repeats the last operation (e.g. 1 × 13 = = = = gives 13^5), and a new number followed by = reuses the last operation and second number as a constant.
%	percent	Percentages, percentage ratios, markup and discount (125 + 3 % → 128,75).
/−/	minus sign	Changes the sign of the displayed number (like ±).
↔	exchange	Swaps the displayed number with the one in the working register, i.e. swaps the two operands.
С (red)	сброс – reset	Clears the operating registers (full clear, like AC). Memory is untouched. Also unlocks the calculator after an overflow.
СХ	сброс X – clear X	Clears only the number being entered (like CE).
П+	память + – memory plus	Adds the displayed number to memory (like M+).
П−	память − – memory minus	Subtracts the displayed number from memory (like M−).
П→Х	memory to X	Shows the memory contents; memory is **kept** (like MR).
СП	память	Shows the memory contents and then **clears** the memory. Use it to read a total, or before starting to make sure memory is empty.
Σ+	accumulate	Works like = and also adds the result to memory. See section 4.
РФЕ	fixed decimals	Press РФЕ then a digit 0–9 to set the number of decimal places shown.
РЕЖ	normal mode	Returns to the floating (natural) decimal point.

Display signs: a dot at the **top** of the leftmost position means overflow – the calculator locks until you press С (memory is kept). A dot at the **bottom** of the leftmost position means the memory is not empty. If memory itself

overflows, press СХ or С, then СП to clear it.

3. Decimal modes: РФЕ and РЕЖ

The manual describes Ф (**РФЕ** on later calculators) as the key that switches the calculator to “giving results with a fixed decimal point”, and Е (**РЕЖ**) as the key for “giving results with a natural decimal point”. The manual does not spell out the abbreviations; Ф almost certainly stands for *фиксированная* (fixed) and Е for *естественная* (natural).

Press	Display	Notes
РФЕ 2	0,00	Two decimal places selected
123,4 ÷ 45,675 =	2,70	Result rounded to 2 places
РЕЖ	2,70169677066228	The full value was kept internally; only the display was rounded

Start-up routine (manual §7): switch on, then press **РЕЖ, СП, С** (normal decimals, empty memory, reset). The calculator is ready when a single 0 appears in the rightmost position after about 10 seconds.

4. Using Σ+ (automatic accumulation)

Σ+ finishes the pending calculation like =, shows the result and adds it to memory, so several separate terms can be worked out and totalled automatically.

- Press **СП** first to empty the memory.
- Enter each term as a normal calculation, finishing it with **Σ+** instead of =.
- Read the total with **СП** (shows and clears) or **П→Х** (shows and keeps).

Example from the manual: 8÷3 + 4÷3 − 5×6 (with РФЕ 3)

Press	What happens	Memory
СП	clears memory	0
8 ÷ 3 Σ+	shows 2,667 and adds it	2,667
4 Σ+	shows 1,333 – the ÷3 is reused – and adds it	4
5 /−/ × 6 Σ+	shows −30 and adds it	−26
СП	shows the total **−26** and clears memory	0

Things to watch for

- **The last operation carries over.** After 8 ÷ 3 Σ+, typing just 4 Σ+ gives 4÷3, not 4. Handy when terms share a divisor or multiplier, but a bare number followed by Σ+ is not simply added – use **П+** for that.
- **Negative terms:** make the number negative with /−/ before the operation (5 /−/ × 6), or use П− for a single number.
- **Memory can be used inside a term.** The manual's second example builds 120 + 6 + 140×0,05 + 140 = 273 in memory (СП 120 П+ 6 П+ 140 × 0,05 + 140 Σ+), then 27300 ÷ СП Σ+ recalls 273, clears memory and adds 100; 5460 Σ+ reuses ÷273 and adds 20; finally 600 + П→Х Σ+ shows **720**.

5. Electronics

The schematic shows a small chip set: two К145-series processors doing all the calculation, display and keyboard work, a clock generator and a flip-flop. The logic is p-channel MOS (PMOS) running from negative supplies (−27 V and −32 V).

Chip (schematic ref.)	Role
К145ВВ7П (D2)	**Lower 8 digits, keyboard and number processing.** Factory description: input of the eight lower digits and preparation for input of the eight upper digits; conversion (multiplication, division, sign change, single decimal point, rounding in fixed-point mode); display output of the eight lower digits and of the number of fixed decimal places; works together with the К145ВВ8П. The keyboard return lines К1 and К2 go only to this chip.

Chip (schematic ref.)	Role
К145ВВ8П (D1)	**Upper 8 digits (positions 9–16).** Partner of the ВВ7. No separate description was found; its role follows from the ВВ7 description and the matching layout on the schematic.
КР165ГФ2 (G1)	Clock generator. Labelled “ГФ2” on the schematic and listed in the parts listing. Supplies the four clock phases F1–F4 to both К145 chips (pins 1–4 on each).
К172ТР1 (D3)	Two-stage (master–slave) PMOS flip-flop with input logic: multiple S/R inputs, separate clocks C1 and C2, both stages' outputs available. −27 V supply, up to 200 kHz, ≤1.2 µs delay, ≤80 mW. Clocked by the F1 and F3 phases, with an RC network (C2, R3) from −27 V on its inputs; its output goes to pin 5 of the clock generator. Probably start-up or timing control for the clock (not confirmed).

How the chips work together

- **One register split across two chips:** each chip's register output (Выход RG) feeds the other's input (Вход RG), so the 16-digit number circulates through both; the sync line (ВВ7 pin 30 to ВВ8 pin 15) keeps them in step. These lines pass through link jumpers on connector E1 (contacts 2–3, 4–5 and 10–11).
- **Display:** the D outputs select display digits and the I outputs drive segments. Each chip drives its own half of the main display HL1; digits 1–8 (right) are the ВВ7, digits 9–16 (left) the ВВ8. A separate single-digit tube, HL2 (probably an ИВ-6), is driven by the ВВ7.
- **Keyboard:** the keys are scanned with D lines from both chips, but the return lines K1 and K2 go only to the ВВ7.
- **Power supply A1:** −32 V (U_{cc2}, pin 39 of both К145 chips), −27 V (U_{cc1}, pin 48; also the clock generator), and ~5 V and ~0.87 V AC, probably for the display filaments.
- **The comma takes a whole digit position,** so a number has at most 16 digits, or 15 plus the comma; the smallest is 0,00000000000001 (10^{-14}).

6. Repair notes

- **Calculator dead or display blank:** check the supply rails (−27 V, −32 V, filament supplies), then the four clock phases on pins 1–4 of either К145 chip, then the E1 links (2–3, 4–5, 10–11). If any link is missing, broken or corroded, the calculator will not work at all.
- Because both К145 chips share the circulating register and the clock, a fault in either one, the clock generator or the К172ТР1 usually stops the whole display, not just one half.
- **Only one half of the display wrong:** check that chip's D and I lines to HL1.
- **Keys not responding but display fine:** check K1 and K2 into ВВ7 pins 29 and 31, diodes VD3 and VD4, and the reed switches.
- PMOS chips are static-sensitive: the К145ВВ7П passport allows no more than 100 V between any two pins, and the К172ТР1 datasheet limits static to 30 V and any positive voltage on a pin to +0.3 V. Use anti-static precautions.
- The commonly available К172ТР1 datasheet (eandc.ru) contains errors: its diagram is captioned К172ЛМ2 (a different chip), and the pin list (pin 8 = supply, pin 6 = Q2) disagrees with the drawing (Q2 on pin 8). Confirm with a second source before applying power.
- The manual says not to open the calculator or work inside it while it is plugged in. Mains voltage is present.
- Full pinouts and chip details are in *Elektronika_MK-59_Chip_Reference_EN.pdf*.

7. Sources

- Микрокалькулятор “Электроника МК-59” – Руководство по эксплуатации (operating manual; scan from leningrad.su/museum).
- МК-59 Схема электрическая принципиальная (circuit diagram, Приложение 3).
- К145ВВ7П factory passport (этикетка) and К172ТР1 datasheet (eandc.ru).
- 155la3.ru – Музей электронных раритетов, К145 series page: www.155la3.ru/k145.htm
- Habr (2024), “Как пользоваться микрокалькулятором”; ru.wikipedia.org; myadel-gimnaz.by; priborazbor.ru.